

Yang Pei

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Education

National University of Singapore

PhD – Computer Science

- Supervisor: Prof. Mike Zheng Shou
- GPA 4.98/5 (5 semesters)

Singapore

Aug 2023 – Current

(Expected Aug 2027 Graduation)

University of Electronic Science and Technology of China

University of Glasgow

BEng – Electronic Information Engineering (with Honours, 4%)

BEng – Electrical and Electronic Engineering (1st Class Honours)

- GPA 3.96/4 (raw 91.52/100), student ranking 2/242
- Sichuan Provincial Outstanding Graduate

Chengdu, China

Sep 2019 – Jun 2023

Publications

Diffusion Safety & Security

RingID: Rethinking Tree-Ring Watermarking for Enhanced Multi-Key Identification

Hai Ci*, [Pei Yang](#)*, Yiren Song*, Mike Zheng Shou

ECCV 2024 [Paper](#) | [Project Page](#) | [Code](#)

Can Simple Averaging Defeat Modern Watermarks?

[Pei Yang](#)*, Hai Ci*, Yiren Song, Mike Zheng Shou

NeurIPS 2024 [Paper](#) | [Presentation](#) | [Code](#)

IDProtector: An Adversarial Noise Encoder to Protect Against ID-Preserving Image Generation

Yiren Song*, [Pei Yang](#)*, Hai Ci, Mike Zheng Shou

CVPR 2025 [Paper](#) | [Code](#)

WmAdapter: Adding Watermark Control to Latent Diffusion Models

Hai Ci, Yiren Song, [Pei Yang](#), Jinheng Xie, Mike Zheng Shou

ICML 2025 [Paper](#)

DiffSeg30k: A Multi-Turn Diffusion Editing Benchmark for Localized AIGC Detection

Hai Ci*, Ziheng Peng*, [Pei Yang](#), Yingxin Xuan, Mike Zheng Shou

2026 [Paper](#)

Continued overleaf

macOSWorld: A Multilingual Interactive Benchmark for GUI Agents

Pei Yang^{*}, Hai Ci^{*}, Mike Zheng Shou

NeurIPS 2025 [Paper](#) | [Presentation](#) | [Project Page](#) | [Code](#)

In-Context Defense in Computer Agents: An Empirical Study

Pei Yang^{*}, Hai Ci^{*}, Mike Zheng Shou

2025 [Paper](#)

VideoGen and Post-Training

X-Humanoid: Robotize Human Videos to Generate Humanoid Videos at Scale

Pei Yang^{*}, Hai Ci^{*}, Yiren Song, Mike Zheng Shou

2025 [Paper](#) | [Project Page](#) | [Presentation](#) | [Post](#)

H2R-Grounder: A Paired-Data-Free Paradigm for Translating Human Interaction Videos into Physically Grounded Robot Videos

Hai Ci^{*}, Xiaokang Liu^{*}, Pei Yang, Yiren Song, Mike Zheng Shou

2025 [Paper](#) | [Project Page](#)

OmniHumannoid: Decoupled Human-to-Robot Video Generation with Paired-Free Adaptation

Yiren Song^{*}, Xiyao Deng^{*}, Pei Yang, Yihan Wang, Mike Zheng Shou

2026 (Under review)

UENR-600K: A Large-Scale Physically Grounded Dataset for Nighttime Video Deraining

Pei Yang^{*}, Hai Ci^{*}, Beibei Lin, Yiren Song, Mike Zheng Shou

2026 (Under review) [Project Page](#)

ActionMap: Robot Policy Learning via Voxel Action Heatmap

Pei Yang^{*}, Hai Ci^{*}, Yanzhe Chen, Qi Lv, Han Cai, Mike Zheng Shou

2026 (Under review)

Relevant Activities

Teaching Assistants

2025	CS4243	Computer Vision & Pattern Recognition	(Prof Amirhassan Monajemi, 40h lecturing)
2024	GEI1000	Computational Thinking	(Prof Leow Wee Kheng, 72h lecturing)
2024	CS1010	Programming Methodology	(Prof Zhao Jin, 20h lecturing)
2023	UESTC3010	Team Design Project and Skills	(Prof Wasim Ahmad)
2022	UESTC1005	Introductory Programming	(Prof Ahmed Zoha)

Honours

Honourable Mentions

- 2025 **NeurIPS 2025 Top Reviewer**
2023 **Sichuan Provincial Outstanding Graduate (四川省优秀本科毕业生)**

Scholarships

- 2022 **Academic (Half-Tuition) Scholarship**, Glasgow College UESTC (6%)
2022 **Outstanding Student Scholarship (1st Class)**, UESTC (10%)
2022 **Suzhou Industrial Zone Scholarship**, UESTC (1%)
2022 **James Watt Scholarship of Innovation**, Glasgow College UESTC (6%)
2021 **Academic (Half-Tuition) Scholarship**, Glasgow College UESTC (6%)
2021 **Outstanding Student Scholarship (1st Class)**, UESTC (10%)
2020 **Wuliangye Scholarship**, UESTC (0.25%)
2020 **Academic (Half-Tuition) Scholarship**, Glasgow College UESTC (6%)
2020 **Outstanding Student Scholarship (1st Class)**, UESTC (10%)
2020 **James Watt Scholarship of Innovation**, Glasgow College UESTC (6%)

Other Activities

- macOSWorld and Rewarding Environment for Computer-Use Agents** Sep 2024
Pei Yang
Tech Share at Tencent Singapore
- Invited Talk: Digital Watermarking in the Diffusion Era** Sep 2024
Pei Yang
School of Computing Media Lunch Talk [Slides](#)
- Tutorial: Video Diffusion Models** Nov 2023
Mike Zheng Shou, Pei Yang, Jay Wu
[Video](#) | [Slides](#)

Skills & Highlights

- Programming** Python, C/C++, MATLAB, AppleScript
- Computing** PBS, Slurm, Singularity
- Technical Skills** PyTorch, TensorFlow, FFmpeg, WSL, AWS EC2, Franka, Claude Code
- Software** Unreal Engine, Blender, VMware, Adobe Ps/Pr/Ae/Au, DaVinci Resolve
- Highlights** I do both theoretical works ([RingID](#)) and engineering works ([macOSWorld](#)).
I am good at doing presentations (see the [X-Humanoid Presentation](#)).
I care about **details**.